EWB COURSES

AI-POWERED CYBERSECURITY

ASSESSMENT REPORT

Assessment 01: Security Foundations

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Batch / Group	Batch 1
Assessment type	Individual work
Total marks	100
Issued	18 October 2026

Purpose

This report responds to the four tasks in EWB Courses Assessment 01. It analyses the fictional account-security and phishing scenarios, the supplied synthetic sign-in logs, and the safety of an AI-generated security answer. Conclusions are limited to the evidence provided in the assessment.

Confidentiality note: This report uses the fictional training evidence supplied in the assessment. No real account credentials or personal account logs are included.

1. Task A — Secure a Student Account (25 marks)

1.1 Risks, assets, CIA impact, controls and limitations

The scenario identifies five distinct risks. CIA refers to confidentiality, integrity and availability.

Risk	Asset at risk	CIA impact	Control	Limitation
Password reused across accounts	Email and learning-portal accounts	Confidentiality: one exposed password may allow access to both accounts.	Use a long, unique password for each account; use a password manager if available.	A password manager needs protection; unique passwords do not prevent every attack.
Unexpected sign-in prompts approved	Account and personal information	Confidentiality and integrity: an unauthorised session may expose or change data.	Deny unexpected prompts and report repeated requests to official IT.	MFA is not foolproof; a user can still approve a prompt by mistake.
Device updates postponed	Laptop, stored files and account sessions	Confidentiality, integrity and availability: known weaknesses may be exploited.	Install supported security updates promptly and restart when required.	Updates do not eliminate every threat and can sometimes create compatibility issues.
Only one project copy	Coursework and project files	Availability: device loss, damage or malware could make the work unavailable.	Keep a second backup in a separate trusted location and test that it opens.	A backup can also be lost or infected if it is not separated or protected.
Friend uses laptop while Mira is signed in	Open accounts and private files	Confidentiality and integrity: data may be viewed or changed without permission.	Lock the screen when away and create a separate standard user account for the friend.	Separate accounts do not protect files that are deliberately shared or poorly permissioned.

1.2 Prioritised actions

Priority 1 — Stop approving unexpected prompts. Deny the request and report repeated prompts because an accidental approval could enable access quickly.

Priority 2 — Replace the reused password. Use a long, unique password for each account, starting with email. If compromise is suspected, use the official site or app and review active sessions.

Priority 3 — Back up project work. Create and test a second copy to reduce the chance of losing the only version. Then install pending updates and secure the laptop with a screen lock and separate user account.

1.3 Authentication versus authorisation

Authentication checks that a person is who they claim to be, for example when Mira signs in using a password and a second factor. **Authorisation** determines what an authenticated user may do, such as view course materials, submit assignments or change portal settings. Passing authentication does not automatically grant permission to every file or function.

2. Task B — Analyse a Suspicious Message (25 marks)

2.1 Five warning signs

The assessment states that the college normally uses campus.example. The supplied addresses are fictional training examples; the analysis does not prove that a real sender or website is malicious.

Evidence phrase	Why it needs scrutiny
“awards@campus-awards.example” and “Reply-To: help@personal-awards.example”	The sender and reply address differ, and neither matches the stated normal college domain. Verify independently.
“FINAL WARNING” / “expires in 20 minutes”	Urgency pressures the recipient to act before checking the claim.
“enter your college password and OTP”	The message asks for credentials and a one-time code. Do not provide these through a message link.
“Do not contact the college office”	Discouraging independent verification is suspicious and prevents the recipient from checking the claim.
“Reply with a photo of your identity card”	An identity document contains sensitive information; the request is not independently verified.

2.2 Safe verification process

1. Do not click the link, reply, provide a password or OTP, or send an identity-card photo. 2. Open the college portal using a saved official bookmark or an address already known to be official. Alternatively, contact the college office using details obtained independently from official college material. 3. Ask whether the scholarship notice is genuine. 4. Report the message through the college’s official reporting channel.

HTTPS encrypts the connection to a website but does not prove that the site is operated by the college or that the message is genuine. A sender can choose a familiar display name, so the name alone does not verify identity.

2.3 If a password and OTP have already been entered

1. Change the affected password immediately through the official portal or app, not the message link. Change it anywhere else it was reused. 2. Contact the college’s official IT/security team and explain what happened without sending the actual password or OTP. 3. Ask IT to revoke active sessions or tokens and check account-recovery details. 4. Review recent sign-ins, messages, profile changes and other activity with the official team. 5. Preserve the suspicious message for reporting and follow IT’s advice.

2.4 What cannot be established from the message alone?

The message alone cannot establish the actual sender, whether a sender account was compromised, or what the linked site would do. Additional evidence could include full email headers, independently verified domain information, confirmation from the college and authorised security logs. The sample addresses are fictional and should not be visited.

3. Task C — Triage a Sign-in Alert (25 marks)

3.1 Four-point timeline

Time (IST)	Observed events	Interpretation — not proven fact
09:00–09:01	Password succeeded from 192.0.2.10; MFA was approved. The lab notes identify this as the usual classroom egress.	Consistent with the usual sign-in pattern, but the log alone does not prove who used the account.
10:14–10:15	Two password attempts from 198.51.100.23 failed.	Could be mistakes or unauthorised guessing. An IP address alone does not identify a person.
10:16–10:17	Password succeeded from the same source; MFA was denied.	The password was accepted, but the reason is unknown. The first MFA request was not approved.
10:18–10:20	MFA was approved; a session was created at 10:20. The student says they approved a prompt to stop repeated requests and did not sign in after 09:05.	The approval may have enabled an unauthorised session. Further checks are needed.

3.2 Likely and alternative explanations

Likely explanation: Someone other than the student may have known the password and repeatedly triggered MFA requests. The student's approval at 10:18 may have allowed a session to be created at 10:20. This is a concern, not a confirmed conclusion.

Alternative explanation: The student may have used a different network or device without recognising the source IP, or the records may need more context. The IP address alone cannot prove an attack or identify a person.

Two checks: (1) Review authorised identity-provider logs for device, browser, session, MFA and recovery events around 10:14–10:20. (2) Ask the student through a trusted channel whether they recognise the device, browser, location and activity, then compare with approved device or network records.

3.3 Three prioritised actions for the authorised IT team

1. Protect the account: Contact the student through an approved channel, reset the password as appropriate, revoke active sessions/tokens and review MFA and recovery settings.

2. Preserve evidence: Export and protect relevant authentication, MFA and session logs with timestamps and access controls. Record actions taken and retain the original evidence.

3. Investigate and monitor: Review account changes and related alerts, monitor for repeated attempts, and escalate through the incident-response process. Do not guess who controlled the IP address.

3.4 Incident summary (60–100 words)

On 18 October 2026, student07 had a successful sign-in from the usual classroom egress at 09:00–09:01. Password attempts from 198.51.100.23 failed at 10:14 and 10:15, then succeeded at 10:16. MFA was denied at 10:17 and approved at 10:18; a session was created at 10:20. The student says they did not sign in after 09:05 but approved a prompt to stop repeated requests. Unauthorised access is a concern, but attribution is uncertain. IT should protect the account and preserve logs immediately.

4. Task D — Review an AI Security Answer (25 marks)

4.1 Five unsafe or unsupported claims/actions

AI claim/action	Correction and reason
“The email is safe because it uses HTTPS.”	HTTPS protects the connection but does not prove that the site or sender is genuine. Verify through an independently obtained official channel.
“The logs prove that the person using 198.51.100.23 stole the password.”	The logs show events associated with an IP, not the identity of a person or how the password was obtained. More evidence is needed.
“MFA prevents every account compromise.”	MFA reduces risk but can be defeated by prompt fatigue, social engineering, stolen sessions or other weaknesses.
“Delete the suspicious logs.”	Logs may be evidence. Preserve them securely and follow the authorised incident-response process.
“Send the student’s password to support so they can test it.”	Never send passwords or OTPs to support. Use the official reset process and share only necessary incident details.

4.2 Improved prompt for an AI tool

> *“Analyse only the fictional email evidence and synthetic sign-in logs supplied in this assessment. Separate directly observed facts from interpretations and assumptions. Identify risks without claiming that an IP address identifies a person or proves a crime. State what cannot be established and suggest additional authorised checks. Recommend safe, prioritised defensive actions. Do not ask for or repeat any real passwords, OTPs, identity documents or personal data. Explain uncertainty and give a reason for each conclusion.”*

4.3 Corrected security advice (100–150 words)

Do not trust a scholarship message only because it uses HTTPS or displays a familiar name. This message has several warning signs: urgent pressure, a different reply address, a request for a password and OTP, discouragement of official contact, and a request for an identity-card photo. Do not open the sample link or send the requested information. Verify the scholarship through the college portal or contact details obtained independently from official college material. If a password and OTP have already been entered, change the password through the official site, change it anywhere else it was reused, and contact the college’s official IT team immediately. Ask the team to review and revoke suspicious sessions and check account activity. Preserve the message and relevant logs. An IP address alone does not identify an attacker. Never send passwords or OTPs to support.

4.4 AI-use declaration and evidence/reference notes

AI-use declaration: AI assistance was used to organise and draft this report in simple English from the supplied assessment. The student should review, understand and edit the report before submission and is responsible for the final work.

Evidence note 1 — Assessment: Task B analysis uses the fictional message in the assessment. Task C analysis uses the supplied synthetic log entries and student statement. The evidence does not establish the identity of the person behind an IP address.

Evidence note 2 — References:

1. CISA, *Secure Our World*, general account-security and phishing guidance: https://www.cisa.gov/secure-our-world
2. NIST, *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile* (AI 600-1), guidance on generative-AI risks and evaluation: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

5. Final submission checklist

- Confirm that the name on the cover is correct.
- Batch/group is marked “Not provided.”
- Review and edit the report so the explanations reflect your own understanding.
- Save the final file as EWB_Cyber_Assessment01_YourName.pdf.
- Keep the report until EWB shares the official Google Form and announces the deadline.